Kaarigar Setu: Trust Score and Network Standing

How the two reputation numbers are calculated, with the formulas and the technology behind them.

Both numbers describe a worker from what **employers** said about the worker's confirmed jobs. They answer different questions and are shown side by side on the worker dashboard.

	Trust score (the worker's "credit score")	**Network standing (PageRank)**
Question it answers	How reliable is this worker, given what employers said?	How strongly does the endorsement graph point at this worker, compared with others?
Range	0 to 100, with a band (Excellent, Good, Fair, Building, No history)	0 to 100, best-connected node = 100
Absolute or relative	Absolute: 82 means the same thing tomorrow	Relative: moves when other workers join
Uses the three ratings (quality, punctuality, conduct)	Yes	No
Needs several employers	Yes (confidence rule)	No
Goes into the signed credential	Yes (the band)	No
Changes the other number	Takes PageRank only as a small input to each *employer's* credibility	Never feeds the trust score

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1. Where the data comes from

1. A worker finishes a job. The employer opens the dashboard and starts a **confirmation**: job role, work date, duration, would hire again (yes / no / not stated), and three ratings from 1 to 5 (quality of work, punctuality and reliability, conduct and honesty).
2. The worker receives a one-time code and gives it to the employer, who enters it to confirm. This proves the worker was present.
3. The **fraud model** (Isolation Forest) scores each confirmation on four features: how often the same address and the same device appear, minutes since the employer's previous confirmation, and a reciprocity ratio. Output: an **anomaly score** *a* between 0 and 1. A high score sends the confirmation to a human moderator and it does not count until approved.
4. Only **approved** confirmations feed the two scores. A moderator ruling of "fraud" removes the confirmation and penalises its sender.

Scores are recalculated (all workers, in one pass) whenever a confirmation is approved, a moderator rules on a flagged one, or an account is erased.

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2. The trust score

2.1 Satisfaction of one confirmation (0 to 1)

Each rating `r` from 1 to 5 is converted to `(r − 1) / 4`. The satisfaction `s` is the weighted average of the answers that were actually given:

```
s = ( 0.40*q + 0.25*p + 0.20*c + 0.15*h ) / (sum of the weights used)

q = (quality − 1)/4        p = (punctuality − 1)/4        c = (conduct − 1)/4
h = 1 if would hire again, 0 if not   (left out if not stated)
```

A confirmation with no ratings at all counts as a neutral 0.55 at half weight.

2.2 Leniency calibration

An employer who rates everyone far above the crowd has a little taken off; one who marks far below has a little added. Only employers with at least 3 rated confirmations are calibrated:

```
offset(employer) = clamp( 0.5 * (employer average − average of all rated feedback), −0.10, +0.10 )
s_calibrated     = clamp( s − offset, 0, 1 )
```

2.3 Employer credibility (0.1 to 1)

How much an employer can be believed:

```
credibility = ( 0.40
              + 0.15 * [employer is verified]
              + 0.20 * min(1, approved confirmations given / 5)
              + 0.25 * network standing of the employer ) * max(0.2, 1 − 0.3 * fraud rulings)
              clamped to [0.1, 1.0]
```

The employer's **network standing** (0.3 to 1) comes from PageRank, see section 3.4.

2.4 Weight of one confirmation

```
w = credibility * recency * engagement * (1 − 0.5 * a)

recency    = max(0.15, e^(−age in days / 365))
engagement = 0.8 if duration unknown, else 0.6 + 0.4 * min(1, days / 30)
a          = anomaly score 0..1 from the fraud model
```

2.5 Combining into the score

```
reputation = ( k*0.60 + sum(w * s) ) / ( k + sum(w) )       k = 1.5   (Bayesian average)
confidence = min(1, number of distinct employers / 3)
raw        = 100 * ( confidence * reputation + (1 − confidence) * 0.25 )
score      = clamp( raw − min(15, 5 * fraud rulings about this worker), 0, 100 )
```

- The **prior 0.60** keeps a worker with little evidence near the middle instead of at an extreme.
- **Confidence** means one employer, however glowing, can never produce a top score. The same employer confirming five times is still one source.
- Worked examples (well-established employer, fresh feedback, all 5 stars): one employer gives about 42; three different employers give about 86; five confirmations from the same employer still give about 42; three employers who all give 2 stars give about 30. Each is a unit test.

2.6 Bands

Score	Band
80 and above	Excellent
65 to 79.99	Good
50 to 64.99	Fair
below 50	Building
no approved confirmation yet	No history (not a bad score)

The band is part of the **signed credential** (`trustBand`), so a verifier sees it and it cannot be edited.

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3. Network standing (PageRank)

This is the plain graph view, and it follows the project's design note.

3.1 The graph

- **Nodes:** every employer (`E<id>`) and every worker (`W<id>`) that appears in an approved confirmation.
- **Edges:** one directed edge **employer to worker** per approved confirmation (an "endorsement").

3.2 Edge weight

```
w = h * e^(-age in days / 365) * (1 - 0.5 * a)          kept between 0.05 and 1

h = 1.0  if the employer would hire the worker again
    0.8  if that was not stated
    0.4  if the employer would not
```

Newer, happier, trustworthy confirmations count more.

3.3 PageRank

```
PR(v) = (1 - d) / N  +  d * SUM over edges u -> v of  PR(u) * w(u,v) / SUM over z of w(u,z)

d = 0.85 (damping factor)       N = number of nodes
```

PageRank is a relative score: a node is important if important nodes point to it. Because it works on the whole graph, a worker's standing depends on everyone else.

3.4 Scale to 0-100

```
standing(v) = 100 * PR(v) / max over all nodes of PR
```

A worker with no confirmation has **no value** (it is not shown as 0).

3.5 How PageRank also reaches the trust score

A second, separate PageRank run uses the same endorsement edges with a weaker **reverse** edge (worker to employer, half the weight, minimum 0.01), so well-connected employers earn standing. An employer's scaled PageRank `S` (0 to 100) becomes `0.3 + 0.7 * S / 100`, which is the 0.25-weighted term inside employer credibility (2.3). The worker's *own* network standing never enters the trust score.

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4. Technology used

Part	Technology	Where
Trust score formula	Pure Java 17 (JDK 21 build), no database or network in the formula so every rule is unit-testable	`trust/ReputationScoring.java` (all tunable numbers in `Params`)
Orchestration	Java service: loads approved feedback, runs calibration, asks for PageRank, scores every worker, stores results, audits	`trust/TrustScoreService.java`
Edge weight for network standing	Java static method	`ReputationScoring.endorsementWeight`
Bands	Java enum	`trust/TrustBand.java`
PageRank	Python 3.11, **networkx** `nx.pagerank(graph, weight="weight")` (damping 0.85), **FastAPI** endpoint `POST /trust-score` with Pydantic validation (weight 0 to 1, up to 50,000 edges)	`kaarigarsetu-ml-service/app/services/graph_engine.py`, `app/routes/trust.py`
Java to Python call	HTTP + JSON with an internal token, Gson	`ml/MlServiceClient.java` (interface `MlGateway`)

Part	Technology	Where
Fraud anomaly score	**scikit-learn** Isolation Forest on four features (`ip_frequency`, `device_frequency`, `minutes_since_previous`, `reciprocity_ratio`), trained on synthetic data	`kaarigarsetu-ml-service/app/services/fraud_detector.py`, `app/training/synthetic_data.py`
Employer to worker adjacency list	Java Collections (in-memory graph)	`trust/TrustGraphService.java`
Storage	PostgreSQL 16: `attestations` (ratings 1 to 5 as `SMALLINT`), `trust_edges` (weights), `workers.trust_score`, `workers.trust_band`, `workers.trust_report` (JSON text, includes `networkStanding`)	`database/migrations/004_*.sql`, DAO classes with `PreparedStatement` only
Display	JSP + JSTL dashboard, `<progress>` bars; the signed credential payload carries the band	`worker-dashboard.jsp`, `credential/CredentialService.java`
Tamper evidence	Hash-chained audit log (database triggers plus an HMAC file chain); credentials signed with RSA-3072	`security/AuditLogger`, `credential/`
Verification	JUnit 5 and Mockito (260 Java tests), pytest (ML service), an 80-check end-to-end script	`src/test/...`, `scripts/e2e.py`

Failure behaviour

- **PageRank service unreachable:** trust scores are still computed, with a neutral employer standing of 0.5 for everyone, and the gap is audited (`TRUST_PAGERANK_UNAVAILABLE`). The network standing value is left out and audited (`NETWORK_STANDING_UNAVAILABLE`). A confirmation is never lost because the ML service was down.
- **Malformed data:** the ML endpoint rejects weights outside (0, 1] and oversize graphs; the Java client validates the response.

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5. What these numbers resist, and what they do not

- **Resists:** one employer inflating a worker (confidence rule); an "always 5 stars" employer (leniency calibration); bursts of confirmations from one source (fraud model, human review, anomaly discount); old feedback dominating (recency); employers caught in fraud (credibility multiplied down, a penalty to the worker for confirmations ruled fraudulent).
- **Does not resist:** a ring of colluding people who each hold real, verified accounts and confirm each other slowly over a long time. The fraud model looks for bursts and same-source patterns, not slow collusion.
- **Network standing caveats:** it is relative (the top worker is always 100), shifts as others join, and ignores the ratings, so it must be read together with the trust score.
- **Fraud model trained on synthetic data only:** the Isolation Forest has never seen real fraud, so its anomaly scores are a reasoned first version.
- **Not yet validated on real data:** the weights (40/25/20/15, 0.60 prior, 3-employer confidence, and the rest) are reasoned defaults that live in one `Params` record for easy tuning. They have not been calibrated against real pilot data.

6. Demo data (made-up accounts)

Worker	Trust score	Trust band	Network standing
Pooja Sharma	82.58	Excellent	100.0
Ramesh Kumar	81.45	Excellent	77.59
Sunita Devi	76.21	Good	69.76
Suresh Yadav	57.55	Fair	52.20
Mohammed Irfan	49.81	Building	52.77
Lakshmi Naidu	48.91	Building	61.48
Anil Verma	39.68	Building	43.24

Worker	Trust score	Trust band	Network standing
Imran Ansari, Deepak Rathore	0.00	No history	none

Lakshmi Naidu shows the difference: her network standing is fairly high because PageRank only sees her approved confirmations, but her trust score is held down by a rejected confirmation and by having only two distinct employers (confidence 0.67).